

Temperature and Top-p in Language Model Generation

Temperature is a parameter that controls the randomness of a language model's output. Lower values (e.g., 0.1–0.3) make the model more deterministic by favoring high-probability word choices, resulting in precise, consistent, and factual responses. Higher temperature values (e.g., 0.8–1.2) increase variability by allowing the model to select from a wider range of possible next words, producing more creative, diverse, and less predictable outputs. In essence, temperature adjusts the balance between stability and creativity in generated text.

Top-p, also known as nucleus sampling, regulates diversity by restricting the model's choices to the smallest set of words whose cumulative probability is at least p . When top-p is low, the model selects only from the most likely options, yielding controlled and focused responses. When top-p is high, the model considers a broader distribution of words, increasing expressiveness and variability. Top-p therefore provides a probabilistic threshold that helps manage the trade-off between diversity and coherence in generated text.